

Qatar-2b

R-1939

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_140128 · created 2026-08-06T14:19:47+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456685.93761 +/- 0.00013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.066
Transit depth (Rp/Rs)^2	0.00014 +/- 0.015 [%]
Orbital Inclination (inc)	88.03 +/- 1.74
Airmass coefficient 1 (a1)	0.2 +/- 0.096
Airmass coefficient 2 (a2)	0.9 +/- 0.32
Scatter in the residuals of the lightcurve fit is	54.26 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	4.5
Transit Duration (day)	0.0479 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.066 vs 0.16327 (deviation 2.46σ)
Duration fit vs published	0.0479 d vs 0.0767 d (deviation 13.7σ)
Mid-time O-C	-0.1 min
Depth SNR	0.0
Residual scatter	54.26 %

Light curve

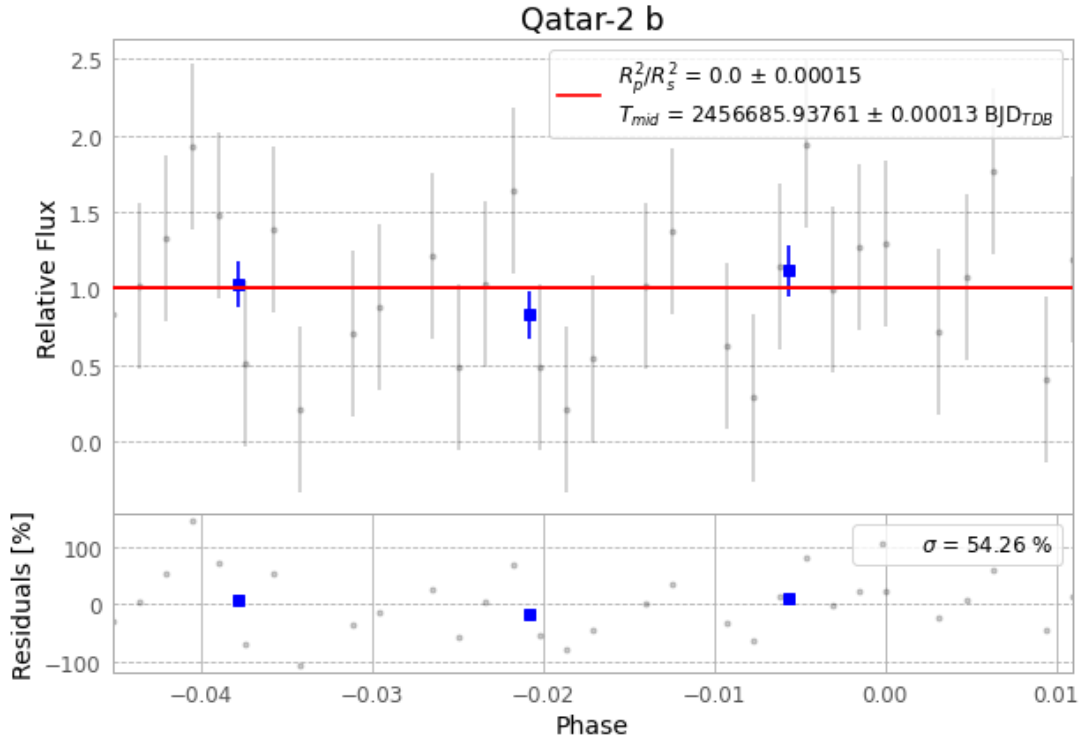


Figure 1 — FinalLightCurve_Qatar-2 b_2014-01-28.png (SHA-256 5738e941d3af674c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [191, 468], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 04a65989cc173fa4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit d8a7143

Data provenance

57 FITS frames (37 MB), Qatar-2140128090012.FITS → Qatar-2140128114812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-140128.log (5e74161ce905...), FinalLightCurve_Qatar-2 b_2014-01-28.png (5738e941d3af...), FinalLightCurve_Qatar-2 b_2014-01-28.csv (58f862785867...)

Compute resources

Wall time 111.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 14:19Z.